

ALGORITHMIC TRADING DECISIONS

As the trading environment has become more complex and competitive, investors have turned to “efficient” algorithms for order execution and navigation. However, utilization of algorithms alone does not guarantee better performance. Investors need to become more proactive than a simple “set and forget” mindset. They need to specify an appropriate set of algorithmic trading rules and corresponding parameters, and most important, ensure that the implementation parameters are consistent with the overall investment objectives of the fund. Otherwise, it is unlikely that best execution will be achieved.

Proper specification of algorithmic parameters requires rules to be specified on a macro- and microscale. The macro-level decisions are specified by users prior to trading and consist of selecting the appropriate optimal trading strategy that is consistent with the investment objectives, and defining how the algorithms are to adapt in real-time to changing market conditions. The micro-level decisions consist of order submission rules and are made at the trade level by the algorithm level. These decisions are made through usage of limit order models and smart order routers.

To ensure “best execution,” investors need to select those brokers and algorithms that can best align the micro-level trading with the user specified macro-level goals.

Macro-Level Strategies

The macro-level strategy decision rules consist of specifying an appropriate optimal trading strategy (e.g., order slicing schedule or percentage of volume rate) and real-time adaptation tactics that will take advantage of real-time market conditions such as liquidity and prices when appropriate. This type of decision making process is consistent with the framework introduced by Kyle (1985), Bertsimas and Lo (1998), Almgren and Chriss (1999, 2000), Kissell and Glantz (2003), and Kissell, Glantz, and Malamut (2004). For investors, macro-level trading specification consists of a three-step process:

1. Choose implementation benchmark
2. Select optimal execution strategy
3. Specify adaptation tactics

To best address these questions, investors need a thorough understanding of market impact, timing risk, and efficient intraday optimization. A detailed explanation of the algorithmic decision making process is provided in Chapter 8.